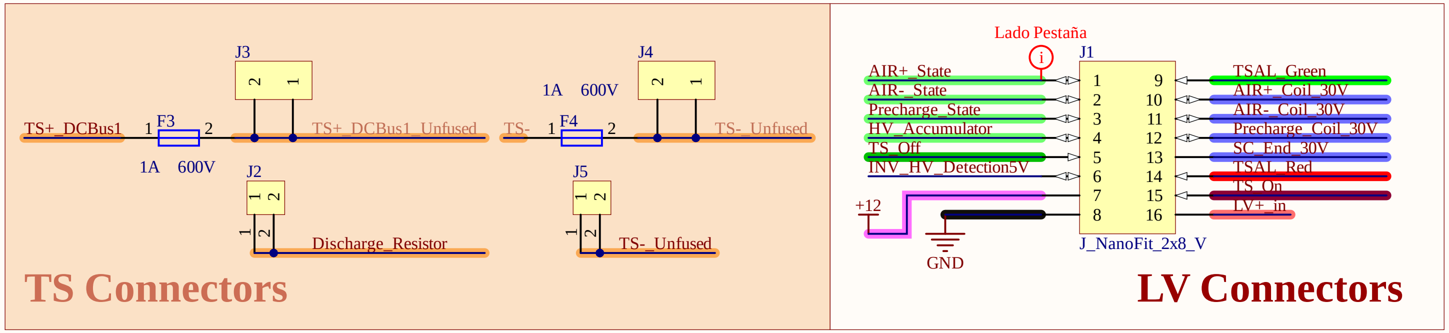
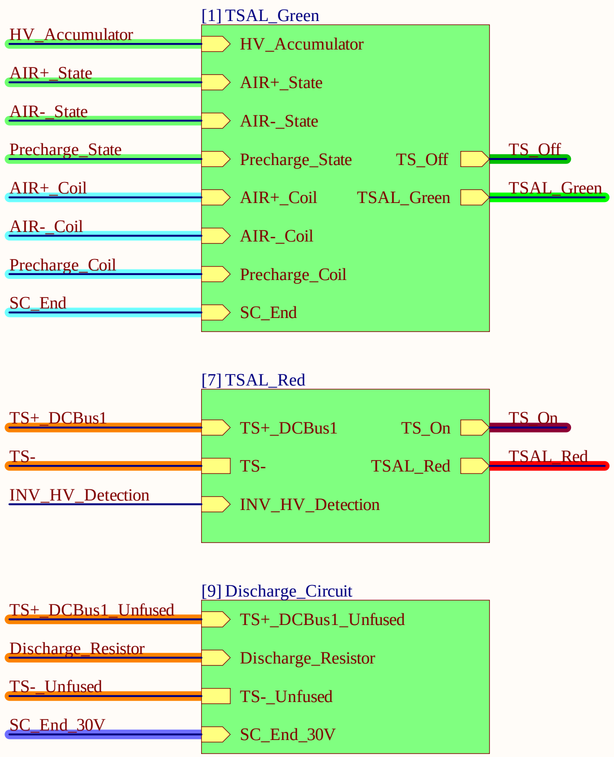
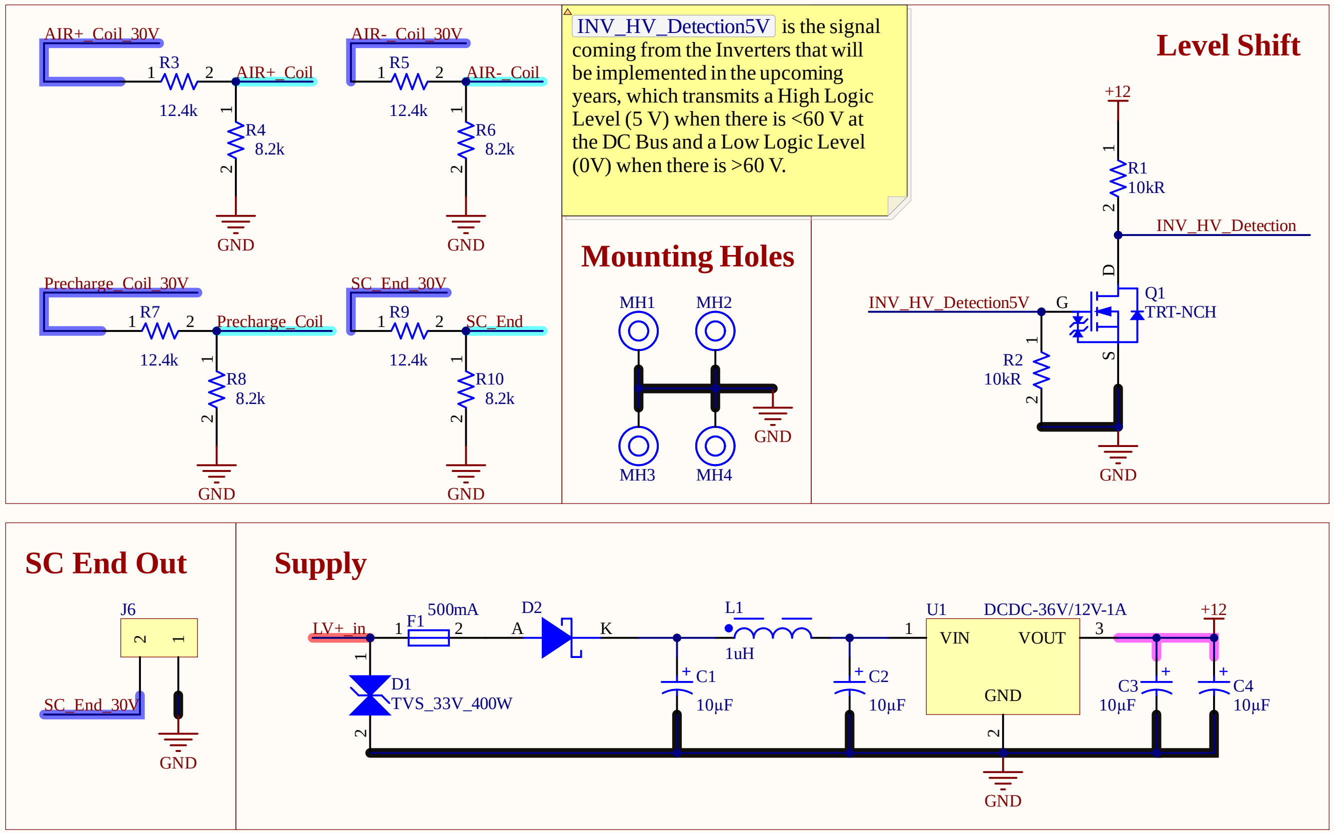
This PCB is installed in the power distribution box (HV Box) and is capable of controlling TSAL_Light according to 2025 FSG rules. It needs signals from AMS_Master (or TSAL_Dummy) and Voltage measurements on the DC Bus.

Signal	12 V	1 V	0 V
AIR+_State	AIR+ Open	AIR+ Closed	Broken wire
AIR-_State	AIR- Open	AIR- Closed	Broken wire
Precharge_State	Precharge Relay Open	Precharge Relay Closed	Broken wire
HV_Accumulator	No HV inside TSAC	HV inside TSAC	Broken wire
Relays_Coil	Relay Open	N/A	Relay Closed







TSAL CONTROL

Title		
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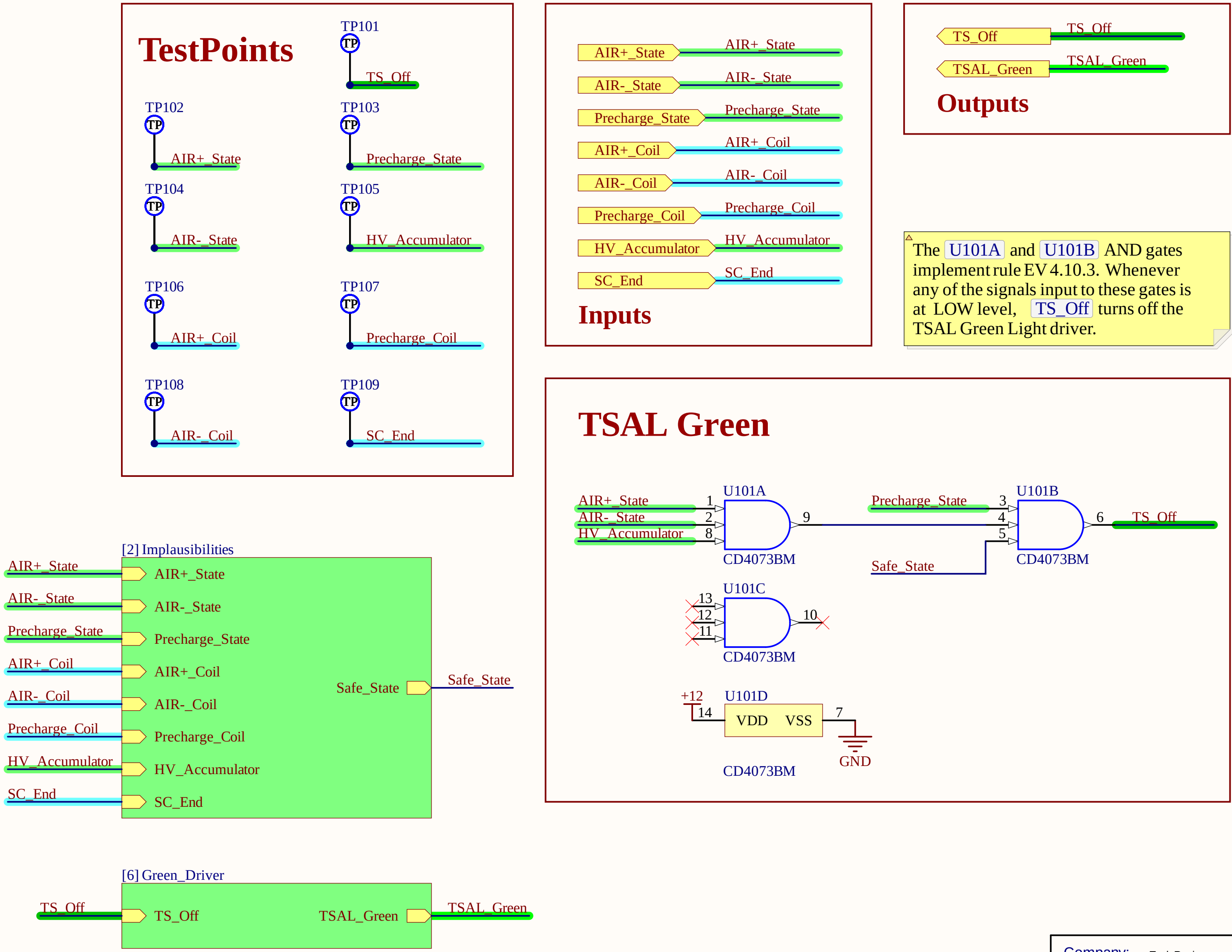
File: E:\Documentos\..\TSAL_Control.SchDoc		Drawn By:



The U101A and U101B AND gates implement rule EV 4.10.3. Whenever any of the signals input to these gates is at LOW level, TS_Off turns off the TSAL Green Light driver.

- AIR+_State : is the signal that indicates the mechanical state of the AIR connected to the positive terminal of the Accumulator.
- AIR-_State : is the signal that indicates the mechanical state of the AIR connected to the negative terminal of the Accumulator.
- HV_Accumulator : indicates if at the vehicle side of TS inside the TSAC there is HV.
- Precharge_State : is the signal that indicates the mechanical state of the Precharge relay connected to the positivr terminal of the Accumulator and the Precharge resistor.

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Project: TSAL Control	Variant: [No Variations]	
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		Department:
Author: Guillem Ropero Serrano guillemroper@gmail.com		Sheet * of *
Checked by:		Date: 12/02/2025

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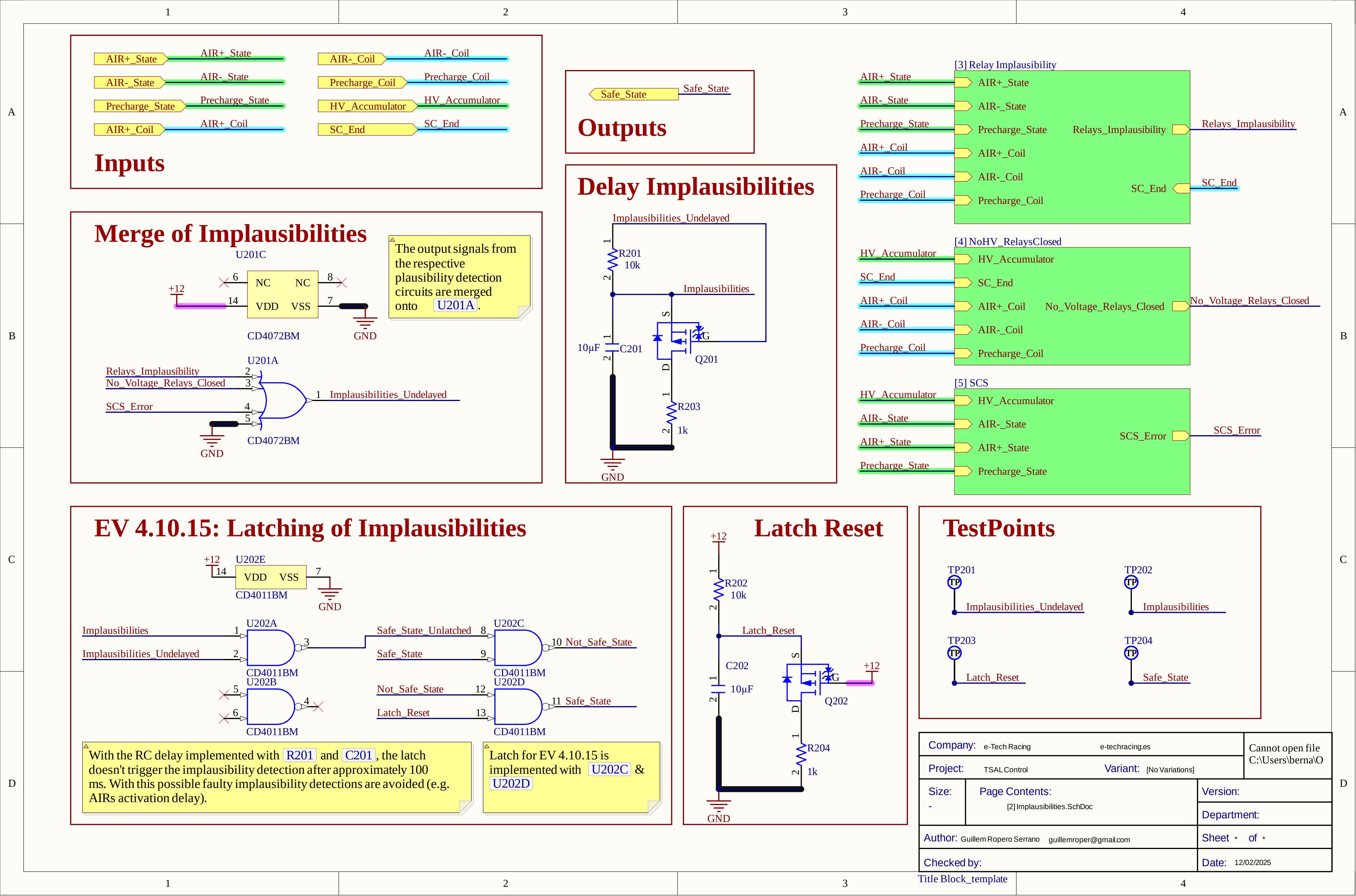

1
2
3
4
A
B
C
D
Inputs
AIR+_State
AIR-_State
Precharge_State
AIR+_Coil
AIR-_Coil
Precharge_Coil
HV_Accumulator
SC_End
Outputs
Safe_State
Merge of Implausibilities
U201C
NC
VDD
VSS
+12
GND
CD4072BM
U201A
Relays_Implausibility
No_Voltage_Relays_Closed
SCS_Error
Implausibilities_Undelayed
The output signals from the respective plausibility detection circuits are merged onto U201A.
Delay Implausibilities
Implausibilities_Undelayed
R201
10k
Implausibilities
10µF
C201
Q201
R203
1k
GND
[3] Relay Implausibility
Relays_Implausibility
SC_End
[4] NoHV_RelaysClosed
No_Voltage_Relays_Closed
[5] SCS
SCS_Error
EV 4.10.15: Latching of Implausibilities
U202E
CD4011BM
U202A
U202B
U202C
U202D
Implausibilities
Implausibilities_Undelayed
Safe_State_Unlatched
Safe_State
Not_Safe_State
Latch_Reset
With the RC delay implemented with R201 and C201, the latch doesn't trigger the implausibility detection after approximately 100 ms. With this possible faulty implausibility detections are avoided (e.g. AIRs activation delay).
Latch for EV 4.10.15 is implemented with U202C & U202D
Latch Reset
+12
R202
10k
Latch_Reset
C202
10µF
Q202
R204
1k
GND
TestPoints
TP201
TP202
TP203
TP204
TP
Implausibilities_Undelayed
Implausibilities
Latch_Reset
Safe_State
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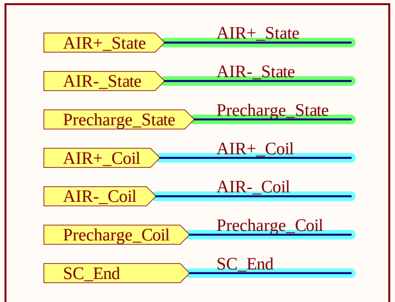
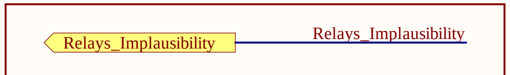


Inputs

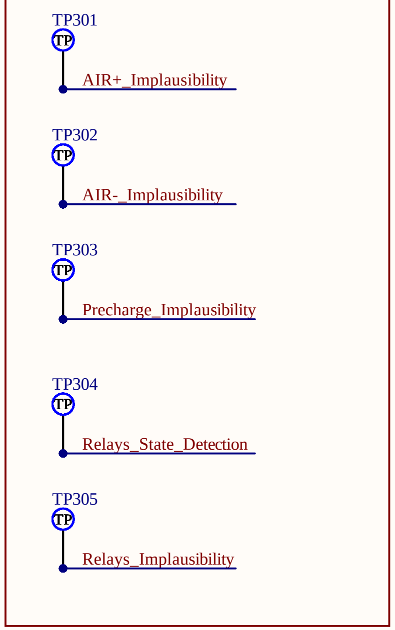


Outputs

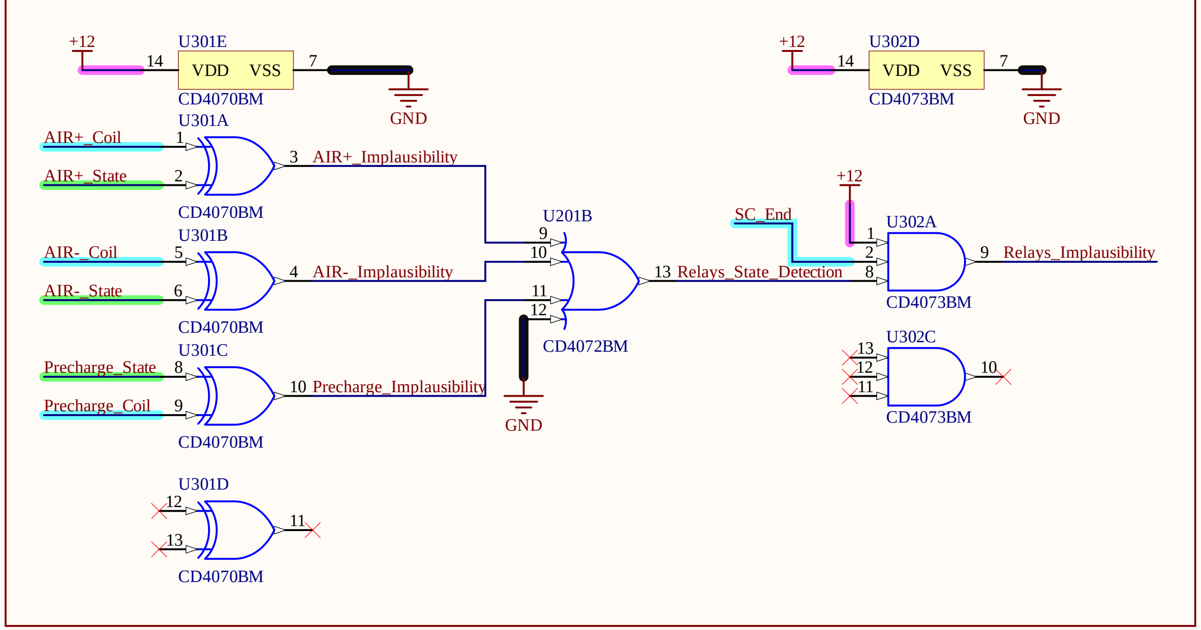
Truth Table

Relay Coil	Relay State	SC End	Implausibility
L	L	L	L
L	L	H	L
L	H	L	L
L	H	H	H
H	L	L	L
H	L	H	H
H	H	L	L
H	H	H	L

TestPoints



Relay State Implausibility



A plausibility check against the mechanical state of the relays is implemented on this sheet. U301A & U301B & U301C & U301D are XOR gates which output a high logic level when the intentional state of any relay (Relay_coil) and the mechanical state (Relay_state) differ. As the Relay's Coil inside the TSAC are supplied directly from the Shutdown Chain, the U302A conditions the implausibility detection with the SC to be closed.

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Project: TSAL Control	Variant: [No Variations]		
Size: -	Page Contents: [3] Relay Implausibility.SchDoc		Version:
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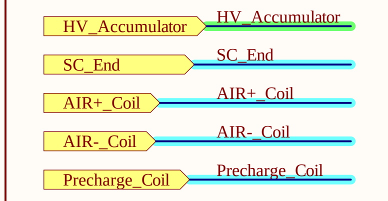
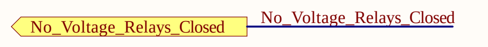
Inputs



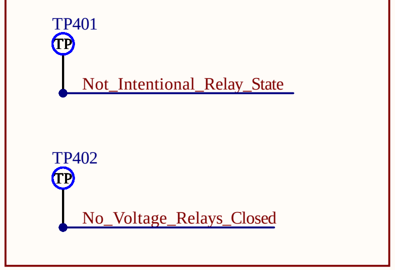
Outputs



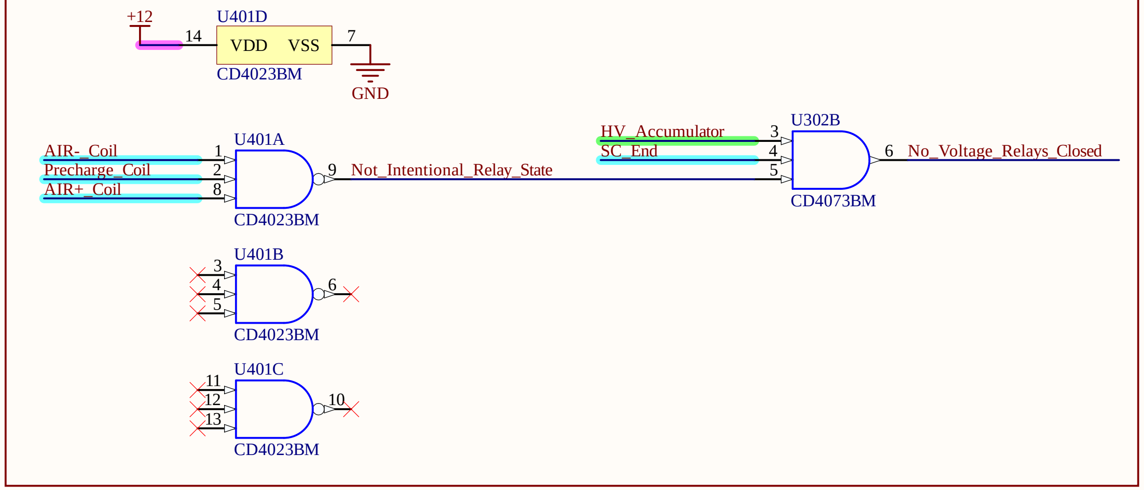
Truth Table

Relay Coil	HV Accum.	Implausibility
L	L	L
L	H	H
H	L	L
H	H	L

TestPoints



EV 4.10.14



In this sheet, it is shown how is implemented a plausibility check against the intentional state of the relays (AIRs + Precharge) inside the TSAC and the presence of HV in the Tractive System inside the TSAC.
When the intentional state of any relay is closed (Low Level), the Shutdown Chain is closed and there is no HV inside the TSAC, Relays_Implausibility shifts to High Level.
As the intentional state of the Relays is taken from the negative pole of the Relay's coil and all the coils of the Relays influencing the TS inside the TSAC are supplied directly from the SC, SC_End_30V coinditions the output of Relays_Implausibility in U302B .

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Project: TSAL Control	Variant: [No Variations]		
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		Department:	
Author: Guillem Ropero Serrano guillemroper@gmail.com		Sheet * of *	
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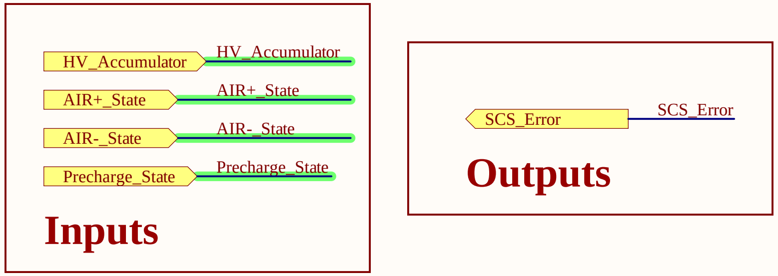
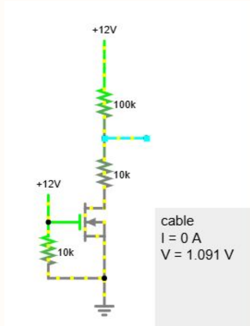
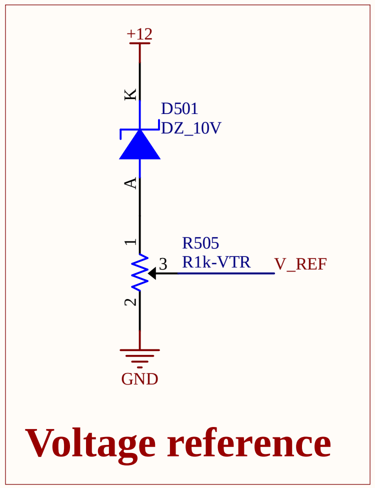
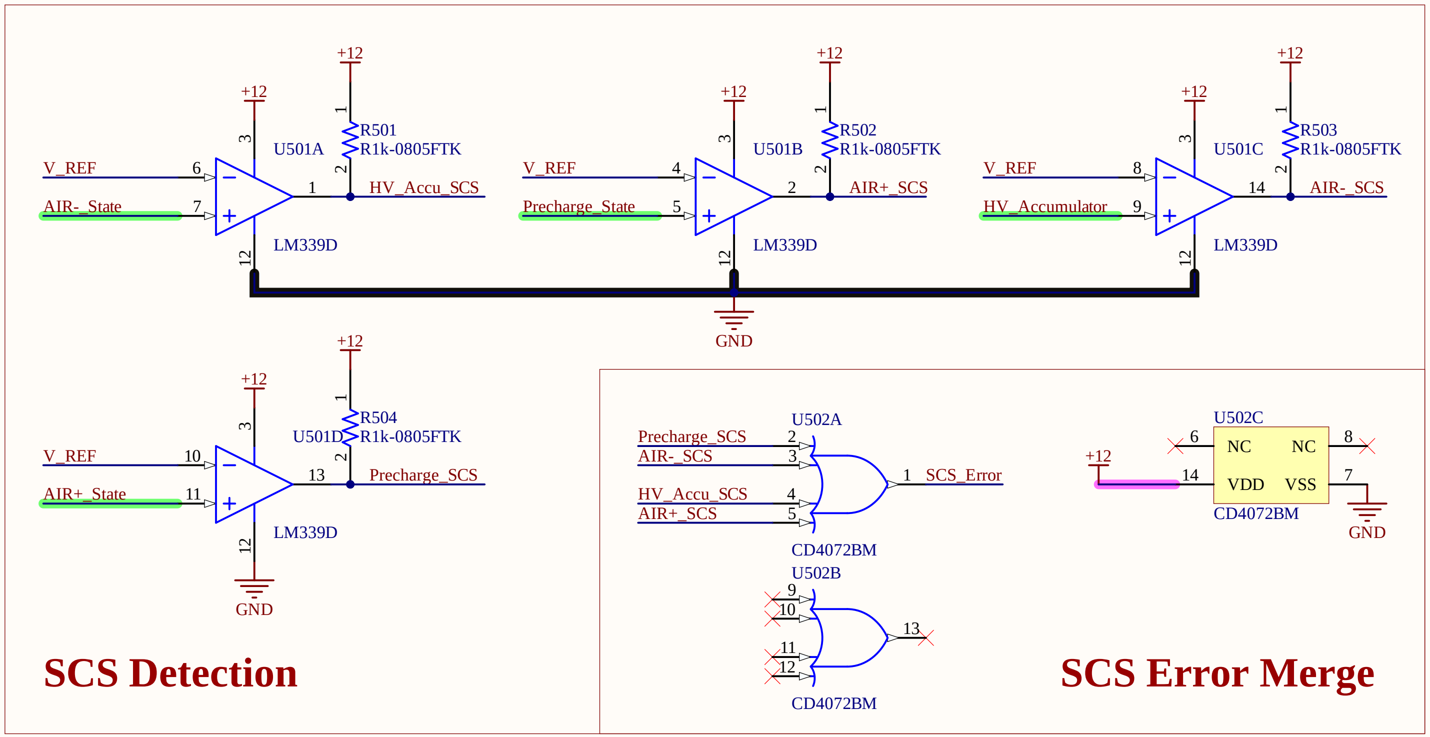
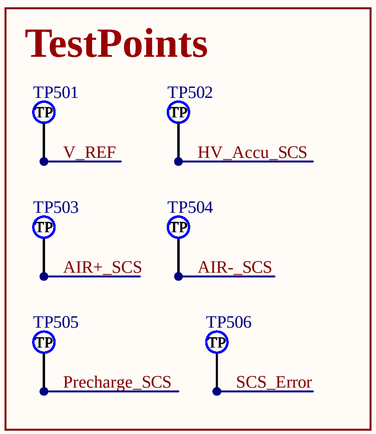




SCS Failure outputs HIGH LEVEL (12 V) whenever any signal from TSAC is at 0 V. Signals coming from the TSAC regarding the mechanical state of the AIRs and Precharge Relay and the HV Detection inside the Accumulator, follow the configuration shown in the image at the left; meaning whenever any of the mentioned signals arrives at 0 V to this PCB there's a broken wire or it doesn't arrive at the correct node.
This is implemented with U501A & U501B & U501C & U501D.







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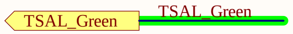
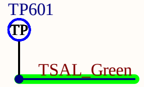
Title Block_template

Inputs

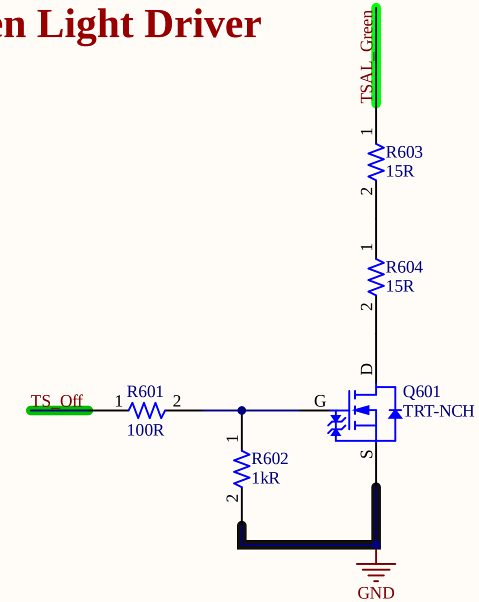
Outputs



TestPoints



TSAL Green Light Driver



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Size: -	Page Contents: [6] Green_Driver.SchDoc	Version:	
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Author: Guillem Ropero Serrano guillemroper@gmail.com		Sheet * of *	
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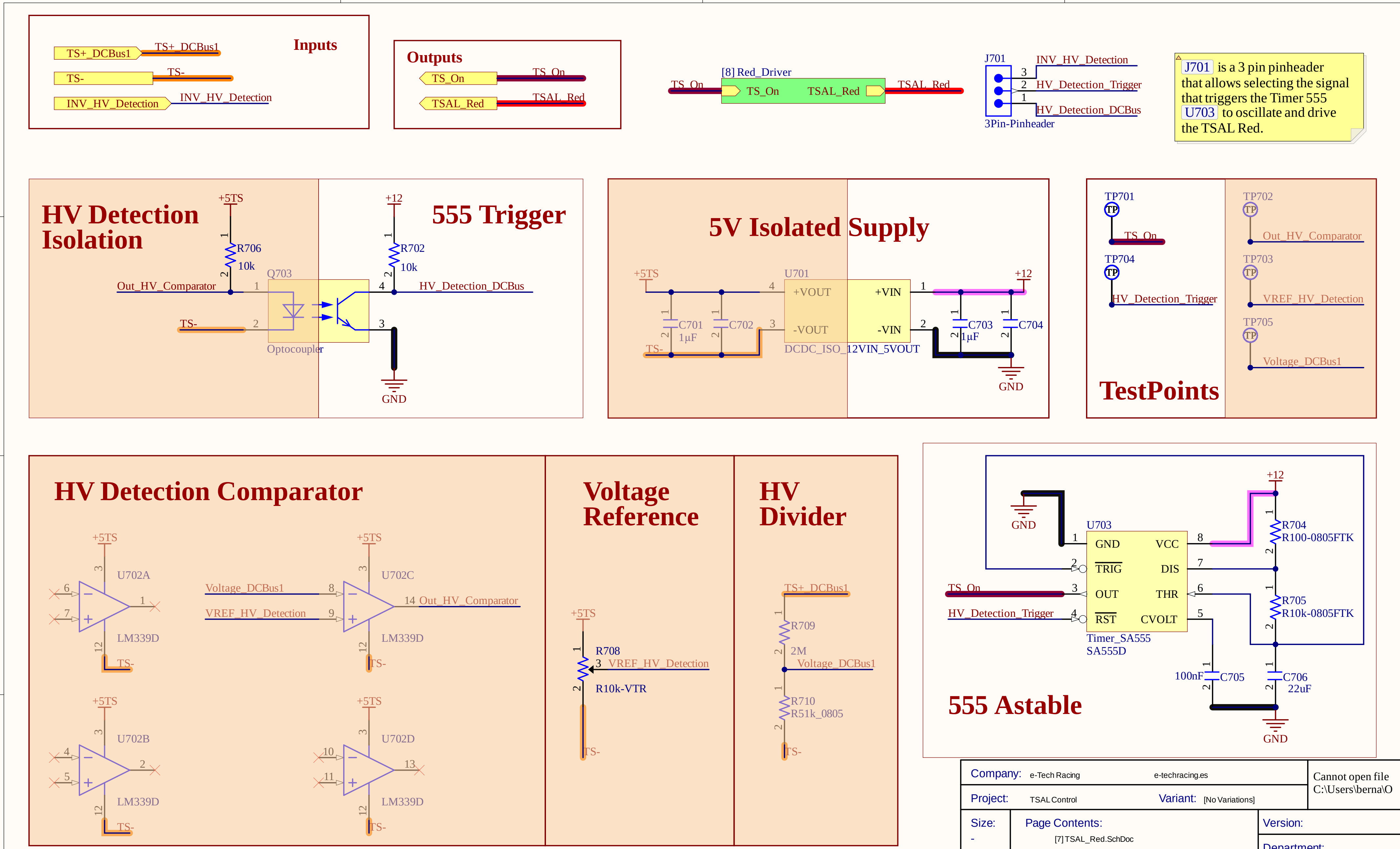

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Project: TSAL Control Variant: [No Variations]

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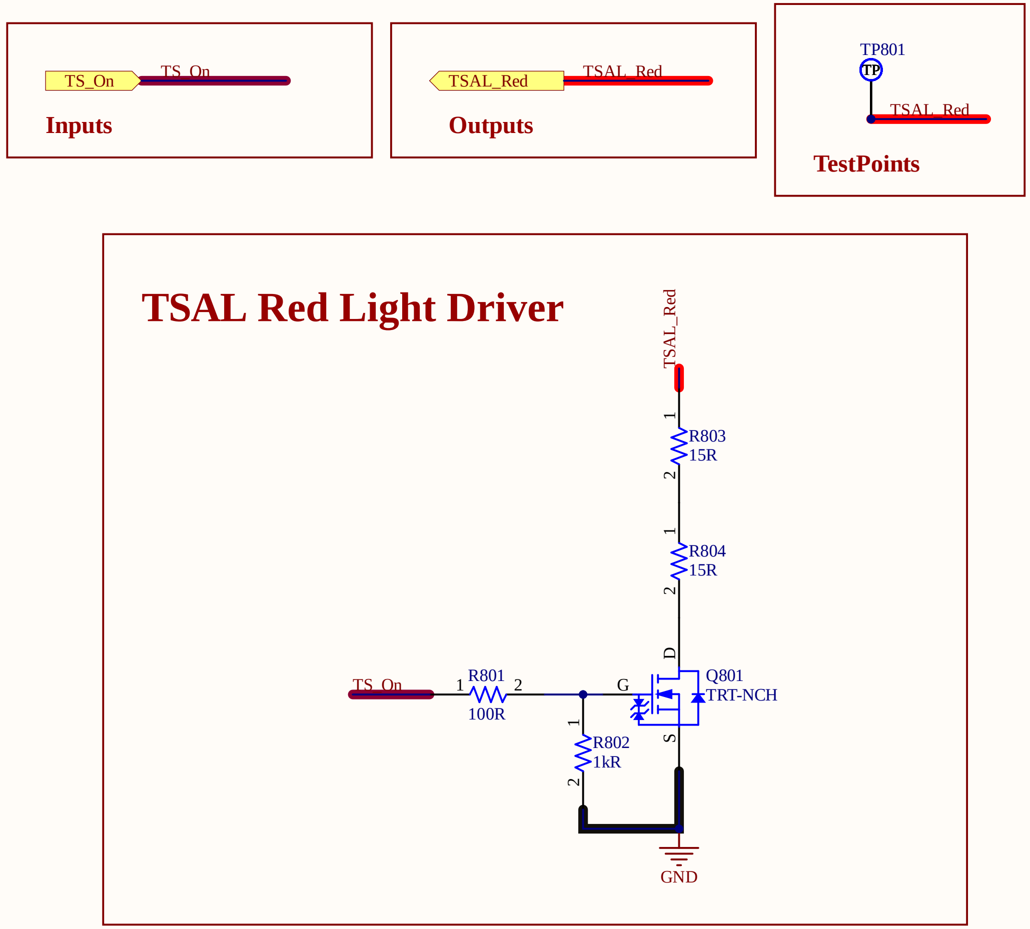

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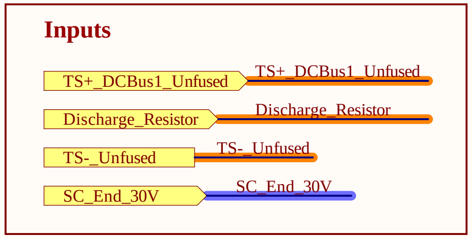
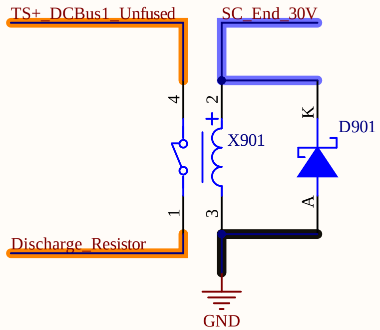


Discharge Relay



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Project: TSAL Control	Variant: [No Variations]		
Size: -	Page Contents: [9] Discharge_Circuit.SchDoc	Version:	
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